

SYNOPSIS

ON

FOOD CALORIE ESTIMATION FROM PHOTOGRAPHS

SUBMITTED IN PARTIAL FULFILLMENT OF THE REQUIREMENT
FOR THE AWARD OF THE DEGREE OF

BACHELOR OF TECHNOLOGY

(Computer Science and Engineering — IoT)

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JAN — MAY, 2026

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Abstract

Accurate dietary assessment is essential for managing health conditions such as obesity, diabetes, and cardiovascular disease. Traditional methods of food logging are manual, tedious, and error-prone, leading to poor adherence among users. This project presents a deep-learning-based system that estimates the caloric and nutritional content of a meal from a single photograph. An EfficientNetV2-S convolutional neural network is trained on the Food-101 benchmark dataset comprising 101,000 images across 101 food categories. Transfer learning is employed in a two-phase training strategy: first, a frozen backbone phase for feature adaptation, followed by fine-tuning of the top layers with cosine annealing learning rate scheduling. The classified food item is mapped to the USDA FoodData Central database to retrieve per-serving nutritional information including calories, protein, fat, carbohydrates, and dietary fibre. A Flask-based web application provides an intuitive interface for image upload and camera capture, returning the top-3 predicted food classes alongside a detailed nutritional breakdown. Experimental results demonstrate a Top-1 validation accuracy exceeding 75% and a Top-5 accuracy above 93%, confirming the viability of photograph-based calorie estimation as a practical tool for dietary monitoring.

Keywords: Food recognition, calorie estimation, deep learning, EfficientNetV2, transfer learning, nutritional analysis, Food-101, convolutional neural network.

1. INTRODUCTION

The global burden of diet-related diseases has intensified the need for accessible and automated nutritional assessment tools. According to the World Health Organization, unhealthy diets and physical inactivity are among the leading risk factors for non-communicable diseases worldwide. Manual food logging through mobile applications, while widely adopted, requires users to search extensive databases and estimate portion sizes — a process that is both time-consuming and subject to significant estimation errors.

Recent advances in computer vision and deep learning have opened new avenues for automating food recognition from photographs. Convolutional neural networks (CNNs), particularly those pre-trained on large-scale image datasets such as ImageNet, have demonstrated remarkable performance in visual recognition tasks through transfer learning. The EfficientNet family of models, introduced by Tan and Le (2019), achieves state-of-the-art accuracy with significantly fewer parameters by employing compound scaling of network depth, width, and resolution.

This project develops an end-to-end system that accepts a food photograph as input, classifies it into one of 101 food categories using an EfficientNetV2-S model, and retrieves the corresponding nutritional information from the USDA FoodData Central database. The system is deployed as a responsive web application built with Flask, enabling real-time dietary analysis through both image upload and live camera capture on any device with a web browser.

2. PROBLEM STATEMENT

Despite the proliferation of diet-tracking applications, accurate food identification and calorie estimation remain significant challenges. The core problems addressed by this project are:

Manual Logging Fatigue: Users abandon food diaries within weeks due to the cognitive burden of searching databases, selecting correct entries, and estimating portion sizes.

Estimation Inaccuracy: Studies report that self-reported calorie intake can deviate by 30–50% from actual consumption, rendering manual logs unreliable for clinical or fitness applications.

Limited Accessibility: Existing image-based solutions often rely on proprietary APIs, cloud-only processing, or require specialised hardware, limiting adoption in resource-constrained settings.

Nutritional Data Gap: Many food recognition systems classify food but do not provide actionable nutritional information, requiring users to perform additional lookups.

Dataset Summary

Parameter	Value
Dataset	Food-101 (ETH Zurich)
Total Images	101,000
Number of Classes	101
Training Images	75,750
Test Images	25,250

3. PROJECT OBJECTIVES

The primary objectives of this project are:

1. To develop a deep learning model capable of classifying food images into 101 categories with a Top-1 accuracy exceeding 75% and Top-5 accuracy exceeding 93%.
2. To implement a two-phase transfer learning strategy using EfficientNetV2-S pre-trained on ImageNet, incorporating frozen backbone training followed by selective fine-tuning.
3. To create a comprehensive nutritional mapping of all 101 food categories using the USDA FoodData Central database, providing per-serving values for calories, protein, fat, carbohydrates, and dietary fibre.
4. To build a responsive Flask-based web application supporting image upload, drag-and-drop, and live camera capture for real-time food recognition and nutritional analysis.
5. To evaluate model performance using standard metrics (Top-1 accuracy, Top-5 accuracy, cross-entropy loss) and validate nutritional estimates against established dietary databases.
6. To design an intuitive user interface that presents classification confidence scores and detailed nutritional breakdowns in a clear, actionable format.

4. DESIGN / METHODOLOGY

The proposed system follows a six-stage pipeline:

Stage 1 — Data Acquisition and Preparation
The Food-101 dataset is downloaded and extracted. Training images undergo data augmentation including random horizontal flips, rotation ($\pm 15°$), colour jitter, and random resized cropping to 224×224 pixels. Test images are centre-cropped and normalised using ImageNet statistics (mean=[0.485, 0.456, 0.406], std=[0.229, 0.224, 0.225]).

Stage 2 — Model Architecture
EfficientNetV2-S pre-trained on ImageNet serves as the feature extraction backbone. The original classifier head (1000 classes) is replaced with a custom head comprising a Dropout layer (p=0.3) followed by a fully connected layer mapping 1,280 features to 101 food classes.

Stage 3 — Phase 1 Training (Frozen Backbone)
The backbone weights are frozen and only the classifier head is trained for 5 epochs using the Adam optimiser with a learning rate of 1×10^{-3} and cross-entropy loss. This phase rapidly adapts the classifier to Food-101 categories without disturbing the pre-trained feature representations.

Stage 4 — Phase 2 Training (Fine-Tuning)
The top 50 layers of the backbone are unfrozen and the entire network is fine-tuned for up to 25 epochs with a reduced learning rate of 1×10^{-4} and cosine annealing scheduling. Early stopping with patience of 5 epochs prevents overfitting.

Stage 5 — Nutritional Mapping
Each of the 101 food classes is mapped to nutritional data sourced from the USDA FoodData Central database. The mapping includes calories, protein (g), total fat (g), carbohydrates (g), dietary fibre (g), and standard serving size per category.

Stage 6 — Web Application Deployment
A Flask web server loads the trained model and exposes a `/predict` API endpoint. The frontend supports drag-and-drop image upload, file selection, and camera capture. Upon submission, the server returns the top-3 predictions with confidence scores and corresponding nutritional information displayed in an interactive dashboard.

5. TOOLS AND TECHNOLOGY

Component	Technology	Purpose
Programming Language	Python 3.9+	Core development language
Deep Learning Framework	PyTorch 2.x	Model training and inference
Pre-trained Model	EfficientNetV2-S	Feature extraction backbone
Dataset	Food-101	101K images, 101 food classes
Nutritional Database	USDA FoodData Central	Calorie and macronutrient data
Web Framework	Flask	REST API and server-side rendering
Frontend	HTML5, CSS3, JS	Responsive user interface
Image Processing	Pillow, torchvision	Image transforms and augmentation
Hardware	Apple M1 Max (MPS)	GPU-accelerated training

6. EXPECTED OUTCOME

1. A trained EfficientNetV2-S model achieving >75% Top-1 accuracy and >93% Top-5 accuracy on the Food-101 test set, demonstrating reliable food classification performance.
2. A comprehensive nutritional database mapping all 101 food categories to USDA-sourced per-serving values (calories, protein, fat, carbohydrates, fibre), enabling immediate dietary feedback upon classification.
3. A fully functional web application (NutriSnap) that processes food photographs in under 3 seconds and presents the top-3 predictions with confidence scores alongside a detailed nutritional breakdown.
4. Validation that two-phase transfer learning (frozen backbone followed by selective fine-tuning with cosine annealing) provides superior convergence compared to single-phase training approaches on the Food-101 benchmark.
5. A responsive, device-agnostic user interface supporting drag-and-drop upload, file selection, and live camera capture, ensuring accessibility across desktops, tablets, and smartphones.

Project Timeline

Phase	Duration	Activities
Phase 1	Jan 2026	Literature review, dataset selection
Phase 2	Feb 2026	Data preparation, model architecture design
Phase 3	Feb–Mar 2026	Model training and hyperparameter tuning
Phase 4	Mar 2026	Web application development and integration
Phase 5	Apr 2026	Testing, evaluation, and optimisation
Phase 6	May 2026	Documentation, synopsis, and final presentation

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